

Making a New Ethernet

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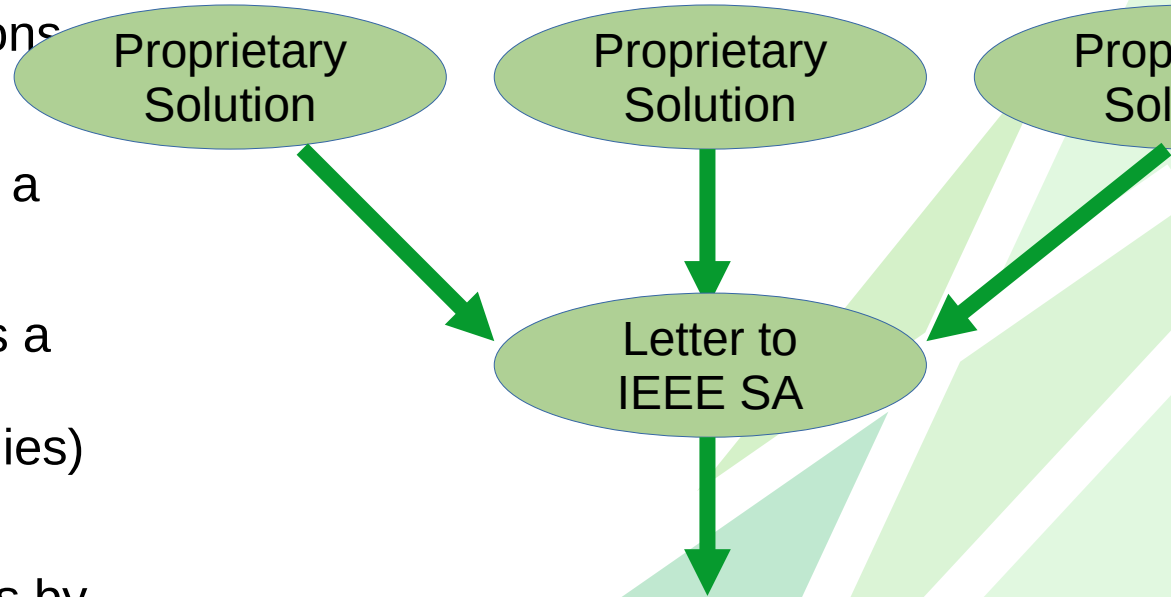
- Right now: Primitive Security. Consulting for FIPS, ESV and hardware security.
- 21 Years at Intel
 - IEEE802 standards development (802.1, 802.3, 802.11, 802.16, 802.21 not in that order)
 - Cryptographic Security
 - Physical Security
 - RNGs & PUFs (RdRand, RdSeed & others)
- Before that
 - Wifi & Bluetooth, F1 Team, Test equipment development, Inmos right out of college.
- Publications – Several Papers, 2 Books
 - Random Number Generators
 - Designing to FIPS140

Who's He?

- The Order of Events
 - Intro
 - Standards
 - Noise!
 - Photons are cheap, or are they?
 - Please Sea Ockra Nose?
 - Link Security

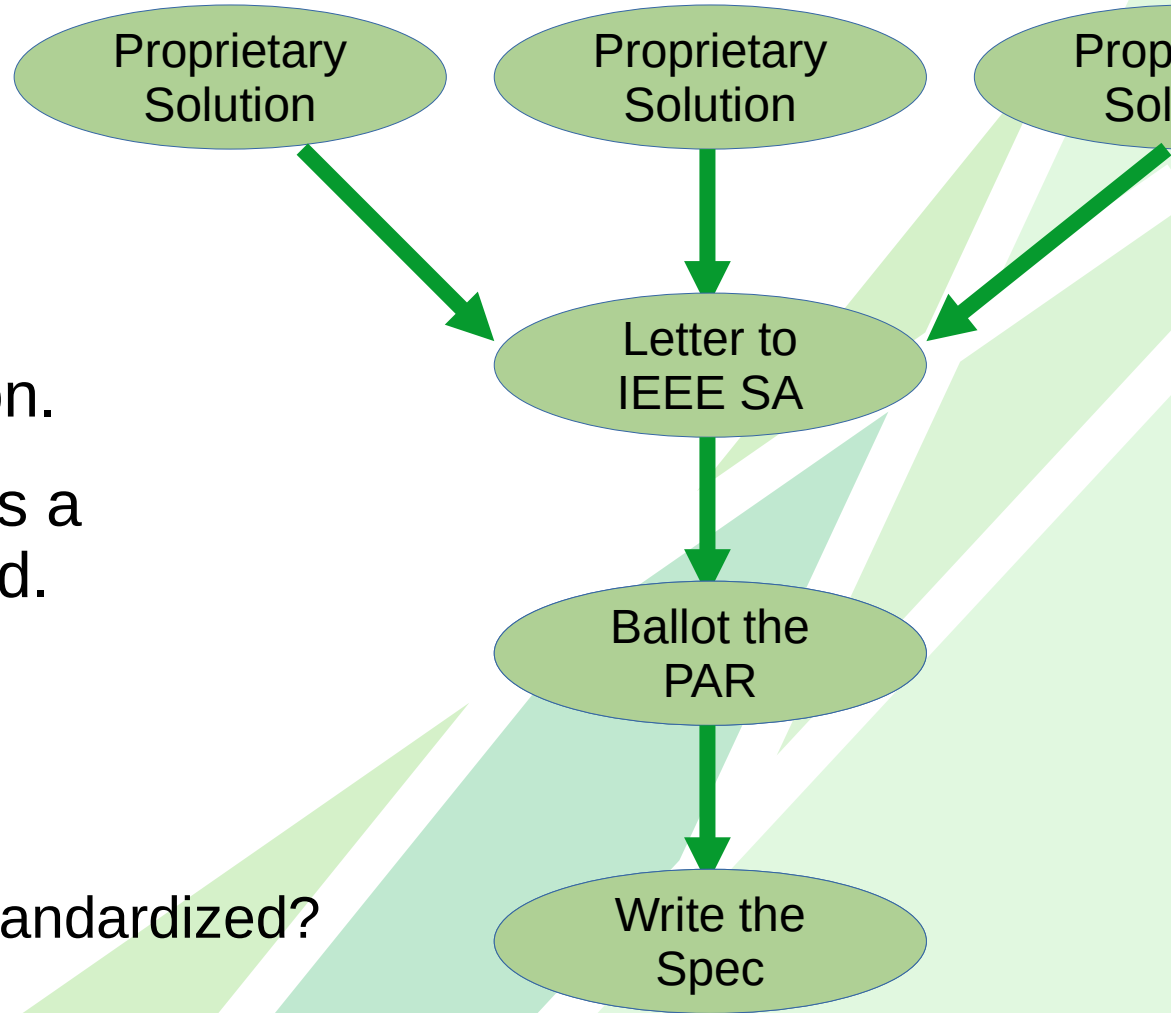
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- First: 1 or more proprietary solutions
- They get together and propose to IEEE SA and P802 that they form a working group (WG) in P802
- Showing broad industry interest is a recipe for success. 802.11 (from several wireless ethernet companies) is an example
- Ethernet would show seriousness by showing interest from ethernet vendors (E.G. cisco, intel, switch companies etc)



How does an 802 medium get standardized?

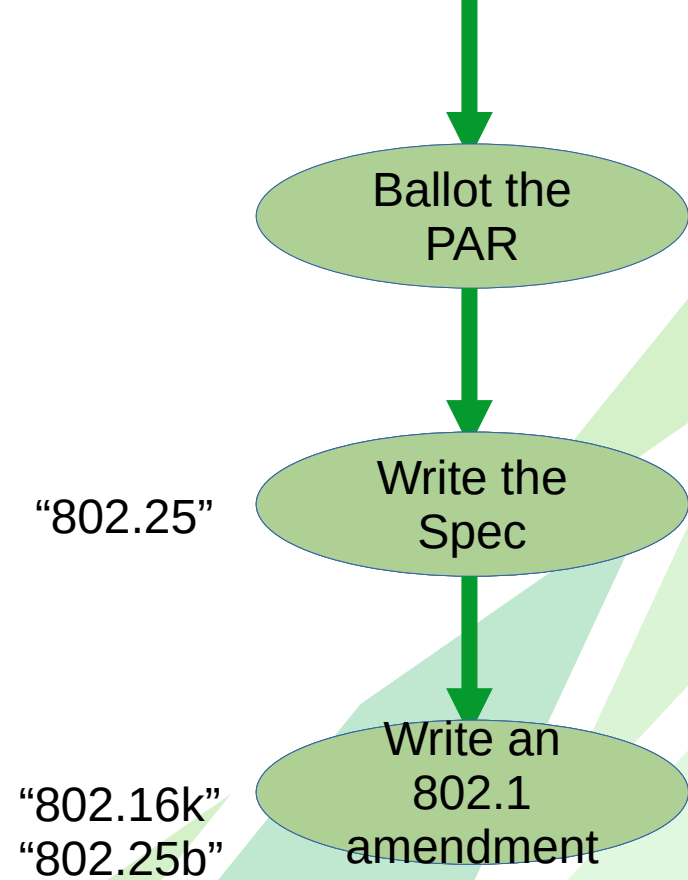
- Form a WG
 - E.G. P802.25
- This needs a PAR documenting scope and purpose. This gets voted on.
- Write your specs. Each has a PAR voted on to get started.



How does an 802 medium get standardized?

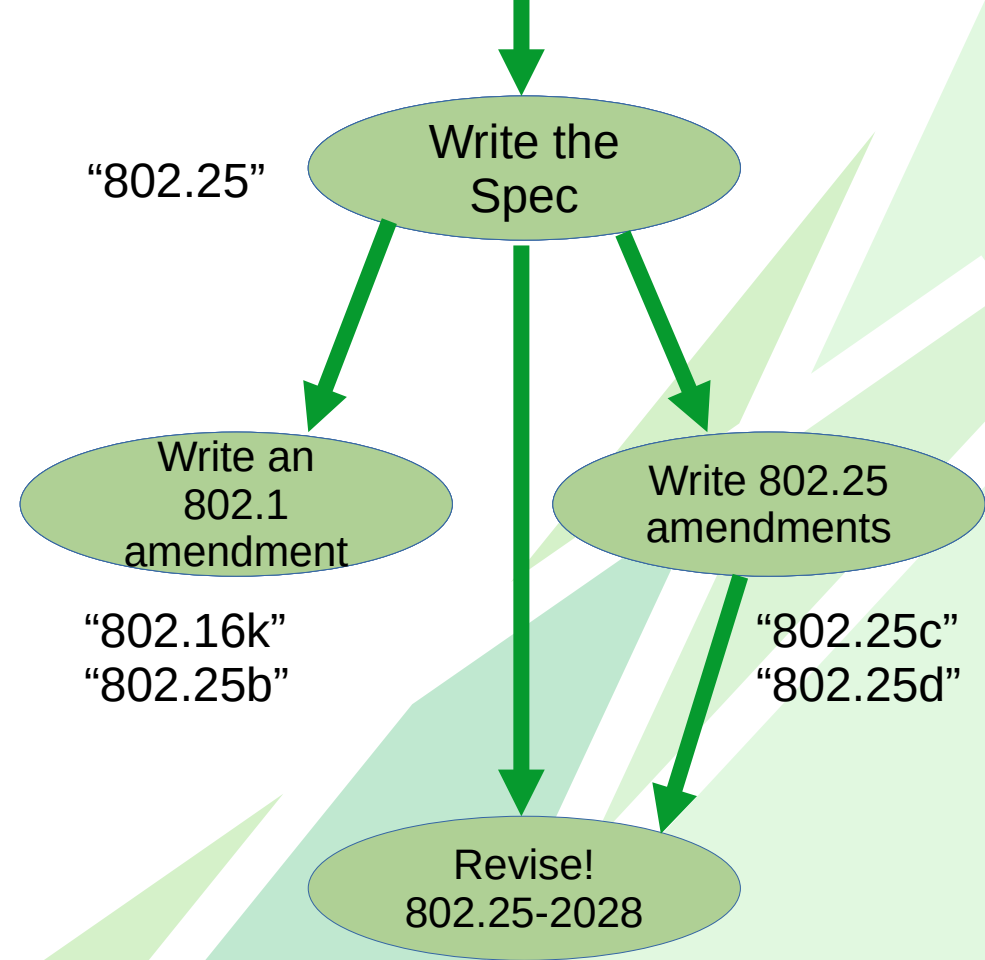
- A medium doesn't just slot into ethernet. If you think it does, you're making assumptions that others won't share.
- Ethernets connect through 802.1 bridges. There are many bridging specs to support many modes of use.
- The bridge-medium interface is the BPDUs-MSDU interface.
- A spec is written in your WG. E.G.. 802.25b. This is not an amendment to 802.25. It's an amendment to 802.1. It covers the encapsulation of BPDUs into MSDUs and visa versa and nails down any assumptions as spec.

So you're close to done?



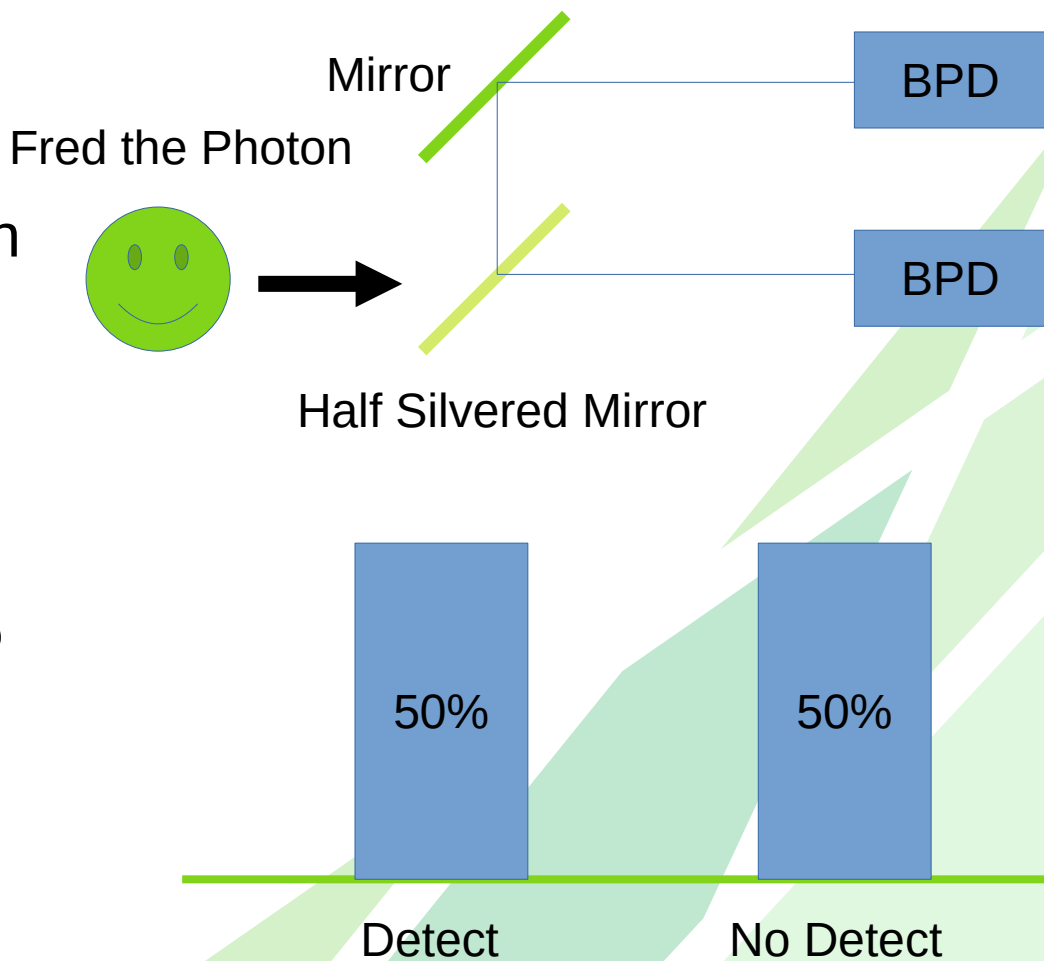
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- Then companies will build ports, test equipment, compliance testing and certification.

What does this buy you?



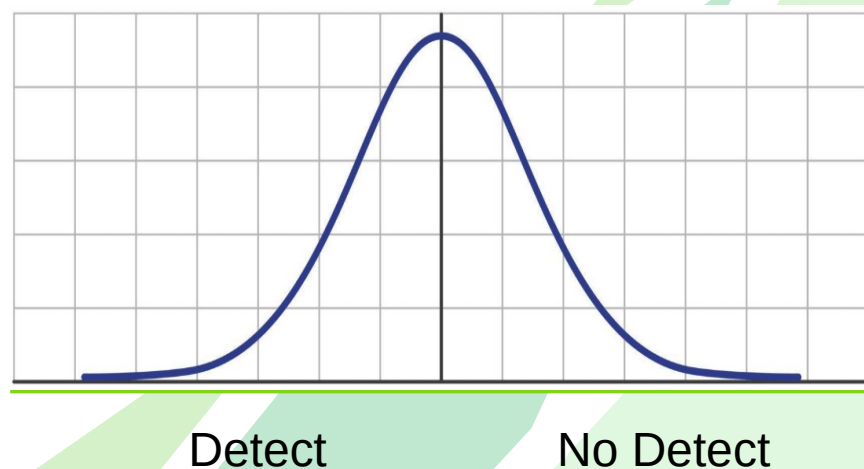
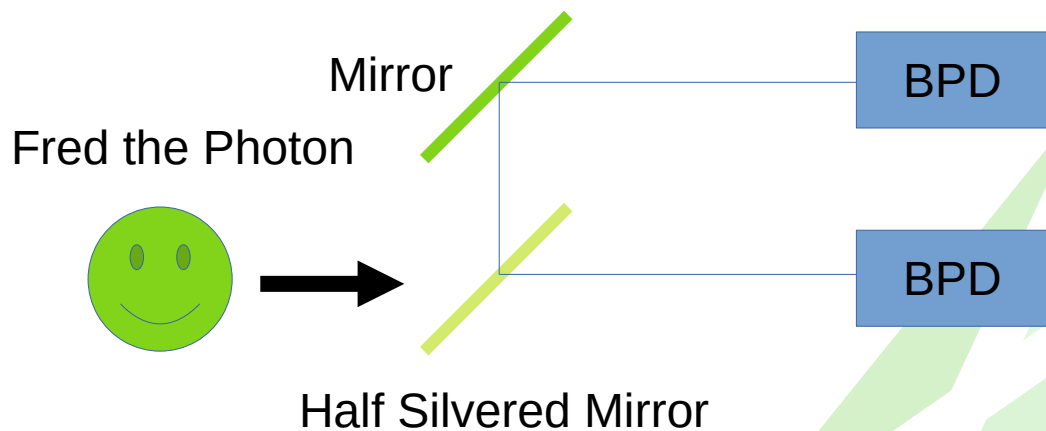
- There's are physics fantasies about the distribution of single photon detectors.

- The mirror is 50% splitting.
- The angle of the mirror is exactly 45%
- The distribution at the BPD is $[1:0.5, 0:0.5]$
- The BPD detects single photons



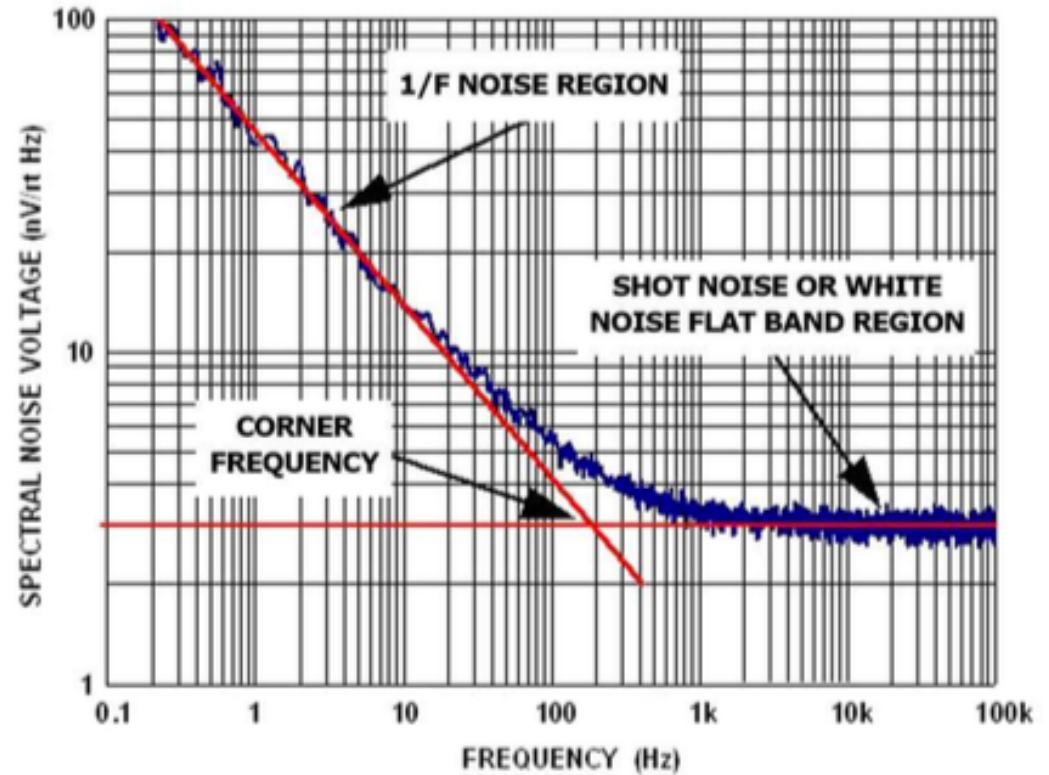
NOISE!

- Single Photon Detectors are not.
- Single Photon Sources output in {nothing, 1 photon, many photons}
- The rest of the environment is entangled with your state
 - Your source, mirror, BPD and atmosphere through which the photon passes is made of atoms
- Light adds. Adding makes Gaussian.

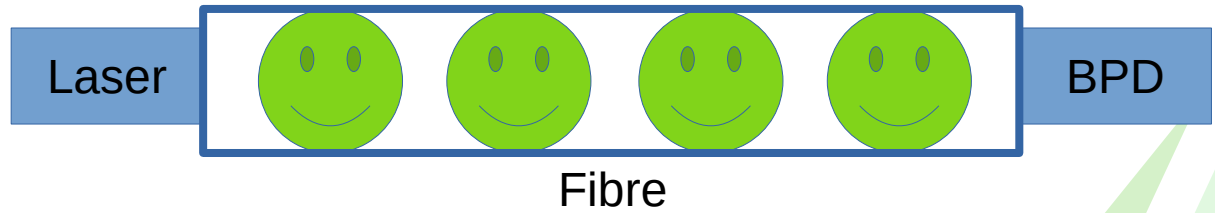


NOISE!

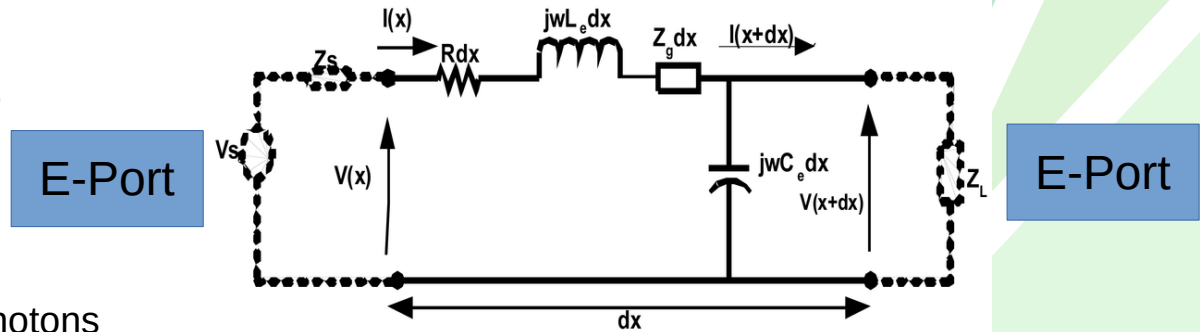
- Not all sources are additive
- Electrons falling in holes
→ $1/F$ Flicker noise
- Small-n Photon arrival (shot noise) – Poisson
- Thermal noise – Check your silicon to how it bubbles up from below



NOISE!



- Photons are easy to make
 - Heat, lasers, LEDs
 - hf is small for each photon
 - Low Energy, lots of photons. Yay!
- Electrical Current
 - Current is easy to modulate
 - Circuits, silicon
 - But electrical current costs
 - $P=VI$ – Much more than photons



Photons are Cheap. Current comes with an electricity bill.

- Crack open your SFP+ router and put your finger on the optical transceiver module.
 - Don't hold it there for long.
 - Pain signals will tell you when to remove it
- Put your finger on a USB ethernet adaptor plugged into your computer.
 - Don't forget to let go.
 - There will be no pain to remind you



Photons are Expensive. Current is Cheap.

- The power is spent in the transceiver
- Wired Ethernet benefits from decades of research into efficient electrical signaling in silicon
- Optical Ethernet has a lower theoretical power requirements but a higher practical power requirement over short distances
- What will this look like in a decade?

What will be cheaper in a decade?

- What's the synchrony model?
- Most fast protocols are source synchronous.
 - At the receiver, it passes data to the local timebase which is asynchronous
 - At the global level, timing specs are adhered to by accurate oscillators but they are not synchronous
 - The global timing assumption is plesiochronous but network timing can pull the global timebase together.
 - Fast, accurate PLLs are the actual backbone of fast protocols
 - E.G. <https://arxiv.org/pdf/2003.05542>

Please See Okra Nose

- Why Link Security?
 - Bridged LAN security is owned by 802.1
 - Routed network security is owned by the IETF
 - Link security is owned by the standardizers of the link.

Link Security

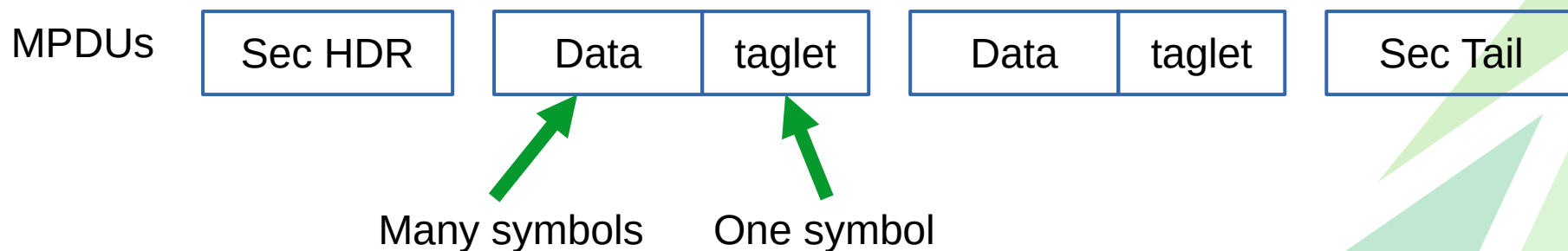
- Authentication costs the same on the wire as authenticated encryption AEAD.
 - Nonce, ICV, header bits
- Link cipher overhead is constant regardless of the packet size.
- Securing lots of tiny packets makes for inefficient link security

Link Security

- Wireless networks have fragmentation.
 - Small packets make ARQ cheap and lightweight
- IP protocols fragment onto 802 ethernet.
 - TLS, IPSec etc. protect IP packets.
 - But small IP packets encrypt less efficiently
- It's ok for a wired ethernet medium to invoke linksec for it's link cipher.
 - It was designed for wired and optical fast links
 - GCM vs CCM

Link Security

- Can we do better?
- Incremental authentication



- Composition Integrity
 - IDs on each endpoint
 - Prove the network topology hasn't changed

Link Security

- Cross Layer optimizations
 - PPDU alignment
 - MSDU/IP alignment



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